HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | LinkedIn | **Stamp 1G** — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed and executed **comprehensive test plans** for mission-critical Java Spring Boot and Python embedded software by analyzing **technical specifications** and collaborating with development teams to define **test scopes and strategies**; validated high-reliability systems across **55% embedded software, 30% backend applications, 10% networking protocols, and 5% web front end** workloads.
- Identified and documented **defects through manual and automated test case execution** across heterogeneous technology stacks (Java, Python, REST APIs, PostgreSQL), performing **root cause analysis** to ensure system reliability in production environments; reduced post-release critical incidents by **40%** through rigorous pre-deployment verification.
- Designed and maintained **automated test scripts** integrated into **CI/CD pipelines** using **Jenkins** and **Git**, ensuring continuous software quality across deployment cycles; automated **regression test suites** covering embedded software and backend services, reducing manual test overhead by **65%**.
- Analyzed **networking protocols** and system interactions between embedded firmware, backend applications, and optical hardware through hardware-in-the-loop testing and integration validation; documented protocol compliance and failure modes to inform design decisions for space-grade reliability.
- Collaborated within **Agile environments** to communicate **test progress and results** directly to developers and product owners using **Jira** ticketing and **Confluence** documentation; coordinated **UAT and post-implementation triage** across interconnected systems, ensuring alignment between testing outcomes and business requirements.
- Developed proficiency in **Software Development Lifecycle (SDLC)** methodologies, **change-control management**, and **problem-solving** for complex system failures; applied meticulous analytical mindset to predict failure modes in high-performance optical systems where rebooting is not an option once deployed in orbit.

SKILLS

Test Planning & Design – test plan development, technical specification analysis, test scope definition, test strategy design, requirements traceability.

Test Execution & Analysis – manual test case execution, defect identification and documentation, root cause analysis, test result reporting, failure mode analysis.

Test Automation & CI/CD – automated test script design, Jenkins pipelines, Git branching, continuous integration, regression test automation, BitBucket Pipelines.

Embedded Systems & Networking – embedded software testing, hardware-in-the-loop testing, networking protocol analysis, backend application testing, web front end testing.

Languages & Frameworks – Python, JavaScript, C, Java, Spring Boot, REST APIs, Flask, FastAPI.

Tools & Platforms – Jira, Confluence, Docker, AWS (EC2, S3, Lambda, CloudWatch), Linux, PostgreSQL, MySQL, Oracle, Splunk, Dynatrace.

Methodologies – Agile/Scrum, Software Development Lifecycle (SDLC), change-control management, UAT coordination, Agile ceremonies.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Free Space Optical Terminal Verification & Test Automation Suite (HCL Technologies)

- Architected comprehensive **test plan** for next-generation FSO terminal embedded software covering **55% embedded firmware, 30% backend microservices, 10% networking protocol stacks, and 5% web UI**; defined **test scopes and verification strategies** for space-grade mission-critical systems by analyzing optical hardware datasheets and firmware specifications.
- Developed and executed **manual test cases** across embedded C/C++ firmware, Python backend services, and JavaScript front-end components; identified **45+ critical defects** including race conditions in hardware-in-the-loop scenarios and protocol compliance violations, performing **root cause analysis** to prevent on-orbit failures.
- Designed **automated test scripts** using Python and **Jenkins CI/CD pipelines** to validate firmware builds, backend API contracts, and networking protocol correctness across every commit; reduced regression testing time from **8 hours manual to 90 minutes automated**, enabling daily deployment cycles.
- Integrated **hardware-in-the-loop (HIL) testing** framework to simulate optical sensor inputs and validate embedded software interactions with physical optical components; discovered **12 timing-sensitive bugs** in firmware that would have caused mission failure in orbit.
- Documented **test results and defect trends** in **Confluence** and tracked issues in **Jira**; communicated verification status directly to development and product teams in daily **Agile stand-ups**, achieving **100% requirement traceability** across test cases and design specifications.

SWIFT Payments Dashboard Test Harness & Operational Monitoring Suite (HCL Technologies)

- Created end-to-end **test automation framework** for SWIFT messaging dashboard supporting **ISO 20022 / MT/MX message flows** and **SEPA cross-border payment** validation; automated **250+ test cases** covering payment-operations workflows, transaction-flow visibility, and regulatory compliance audit trails.
- Executed **UAT coordination** and post-implementation triage with banking-domain operations teams; validated **GDPR data protection** controls and regulatory compliance mechanisms, ensuring **100% traceability** to financial-services SLA requirements and audit-log integrity.
- Designed **monitoring and alerting verification tests** to validate operational dashboard accuracy for payment message tracking; reduced manual monitoring overhead by **70%** through automated P1/P2 incident detection, ensuring mission-critical transaction visibility.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)